Modeling Emerging Market Firms' Competitive Retail Distribution Strategies

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Abstract

In emerging markets, the effective implementation of distribution strategies is challenged by the underdeveloped infrastructure of roads, and a low penetration of retail stores that are insufficient in meeting customer needs. In addition, products are typically distributed in multiple forms through multiple retail channels. Further, due to the competitive landscape, manufacturers' distribution strategies should be based on anticipation of competitor reactions. Accordingly, we develop a manufacturer-level competition model to study the *distribution* and *price* decisions of insecticide manufacturers competing across *multiple product forms* and *retail channels*. Our study shows that both consumer preferences and estimated production and distribution costs vary across brands, product forms, and retail channels; that ignoring distribution and solely focusing on price competition results in up to a 55% overestimation of manufacturer profit margins; and that observed pricing and distribution patterns support competition rather than collusion among manufacturers. Through counterfactual studies, we show that manufacturers respond to decreases in distribution costs and to the exclusive distribution of more preferred manufacturer by asymmetrically changing their price and distribution decisions across different retail channels.

Keywords: retail distribution, emerging markets, manufacturer competition, multi-product form competition, multi-channel competition

Introduction

In most product markets, manufacturers rely heavily on independently owned retail distribution channels to reach their customers. For example, in the U.S. in 2016, there were about 3.8 million retailers whose total retail sales were about $2.6 trillion[1]. In India, there are currently over 14 million retail stores, whose total retail sales are expected to reach $0.95 trillion in 2018 (compared to $0.5 trillion in 2013)[2]. Due to the size and growth of retail sales, product manufacturers consistently make significant monetary efforts to manage their retail distribution channels to make their products accessible, and ultimately to achieve profitability[3]. For example, in India, manufacturers' cost of retail distribution can be as high as 18 to 25% of their total sales[4]. Given the importance of retail distribution for sales, the development of appropriate retail distribution strategies has attained significant importance in the marketing literature (Cao and Li 2015; Kumar et al. 2015; Padmanabhan and Png 1997; Sharma and Mehrotra 2007; Trivedi 1998).

Even though retail distribution is important for product manufacturers to satisfy their customers' needs and ultimately achieve profitability, it is challenging for manufacturers to effectively manage the distribution of their products for a number of reasons. First of all, manufacturers typically manage multiple retail channels that differ in terms of their sizes, target markets, and effectiveness in selling products (Dawar and Frost 1999; Venkatesan et al. 2015). In addition, customers differ in preference with respect to distribution channels depending on the availability of their preferred products in those channels, and on their proximity to those channels. Second, manufacturers typically distribute their products in multiple forms (such as liquid, solid, frozen, etc.) to meet various customer needs (Bloch 1995; Porter and Heppelmann 2015). Thus,

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[1] https://www.selectusa.gov/retail-services-industry-united-states
[2] https://assets.kpmg.com/content/dam/kpmg/pdf/2014/11/BBG-Retail.pdf
[3] https://www.warehousingandfulfillment.com/warehousing-and-fulfillment-resources/the-future-of-distribution-is-promising-for-emerging-markets/
[4] https://www.atkearney.com/consumer-goods/article?/a/rethinking-consumer-product-distribution-in-developing-countries

manufacturers[5] need to decide which product forms to distribute through how many stores in each utilized retail channel. This decision is complicated by the fact that multiple competing manufacturers distribute similar products through common retail channels. To be effective, a manufacturer's retail distribution decisions need to incorporate its competitors' reactions.

Manufacturers' retail distribution decisions can be more complicated in emerging markets for multiple reasons. First, emerging markets typically feature larger underdeveloped infrastructure (such as a lack of highways and roads to reach different regions), and extended geographies with vastly different retail stores that are limited in number (i.e., an insufficient retail landscape)[6]. As a result, it can be difficult for manufacturers to effectively access each of these regions and sales territories through retail distribution (Roberts et al. 2015; Sheth 2011). Second, customers' price sensitivities and preferences toward products have changed widely due to economic liberalization, which has increased the buying power of households in these markets. As a result, even households at the lowest income levels have started to afford a large variety of products[7]. As such, manufacturers have introduced several product forms to serve various customer needs. However, due to the abovementioned challenges (underdeveloped infrastructure and insufficient retail landscape), manufacturers have had difficulties in meeting customers' needs. Third, the penetration of retail distribution is quite low in emerging markets (Mulky 2013). For example, India has the highest number of retail outlets (with an average size of 50 to 100 square feet) worldwide, even though the per capita retail space is the lowest in the world (Pick and Müller 2011). As such, the demand for retail distribution space in India is expected to increase at the rate of 81% in 2018[8]. Similarly, China's trade through retail distribution has increased by 9.7% in

[5]We use manufacturers, brands, and firms interchangeably throughout the rest of the paper.
[6]http://www.mckinsey.com/business-functions/marketing-and-sales/our-insights/building-brands-in-emerging-markets
[7] https://www.strategy-business.com/article/16583?gko=68e98
[8] https://www.ibef.org/industry/retail-india.aspx

2018 over the previous year[9]. Such growth trends in emerging markets create opportunities for firms to increase profitability through appropriate penetration of retail channels. Fourth, manufacturers typically consider retail distribution as a utilitarian task[10]. This commonly results in a retail distribution mechanism that is poorly aligned with customer and market needs (Mulky 2013; Sheth 2011). To overcome all of these challenges specific to emerging markets, emerging market manufacturers need to develop effective retail distribution strategies. In addition, due to the significant growth of emerging economies, multiple firms have ventured into these markets, and therefore the competitive intensity across product forms and retail channels has changed[11]. This leads to the need for understanding retail distribution from a competition perspective in order to maximize profits.

Although analysis of firm competition in terms of price has received considerable attention in the marketing literature, especially in the developed market context (Karray and Martín-Herrán 2009; Sudhir 2001b; Vilcassim et al. 1999), to the best of our knowledge, there is no study modeling emerging market firms' competition in retail distribution. As discussed earlier, since firms typically sell products in multiple forms (e.g., liquid versus solid soap) through various retail channels (e.g., mom-and-pop stores versus big-box stores such as "Big Bazar"), firms' retail distribution strategies should be studied at the *brand, product form,* and *retail channel* levels. Hence, in this study, we investigate firm competition in *retail distribution* along with *price* in an emerging market setting by developing a *multi-product form* and *multi-channel* competition model.

[9] https://www.businesswire.com/news/home/20180411005801/en/Annual-In-depth-Analysis-Report-China-Wholesale-Retail
[10] https://www.strategy-business.com/article/16583?gko=68e98
[11] https://www.strategy-business.com/article/16583?gko=68e98

To model emerging market firms' competition in distribution and price, we use data of competing insecticide manufacturers in India. In the data, we observe: 1) prices at the firm and product form levels (in our setting, liquid and solid product forms); 2) retail distribution levels – or the total number of retail stores through which each firm sells their products – at the product form and retail channel levels (in our setting, paan-plus and general stores[12]); and 3) unit quantity sales (at the manufacturer shipment level) at the brand, product form, and retail channel levels[13] for a period of 48 months. To estimate our supply-side competition model, we use a two-step approach. In our first step, we model the aggregate brand demand (i.e., sales) across different product forms and retail channels using a nested logit (NL henceforth) specification[14] by incorporating unobserved heterogeneity across households in the market and accounting for endogeneity (Petrin and Train 2010). In our second step, we use the estimated aggregate sales model as an input and estimate our supply-side multi-product form, multi-channel manufacturer competition model.

Our study highlights the following results. On the demand side, in terms of the intrinsic preferences, we find that the first manufacturer (Firm 1) in the study, the solid product form, and paan-plus stores are, on average, preferred to the second manufacturer (Firm 2), the liquid product form, and general stores, respectively. On the supply side, we find that the estimated costs of production (distribution) significantly differ not only across firms and product forms (and retail

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[12] We note that both channels in our context are physical distribution channels. We did not consider online channels because they are not prevalent in our setting. Furthermore, the product forms we study are not sold through online channels. This is to note that online retailing has taken off in the Indian market since 2006-2007 (http://www.ey.com/Publication/vwLUAssets/Rebirth_of_e-Commerce_in_India/$file/ey_re-birth_of_ecommerce.pdf; pg. 13). Further, the growth of eCommerce (including e-tailing) in the Indian market was only 3.8% by 2009 (https://www.pwc.in/assets/pdfs/publications/2015/ecommerce-in-india-accelerating-growth.pdf; pg. 5).

[13] In addition, we acquire time-variant information about the average expected target population served by each store in each channel. We use this information to create the unique distribution measure that allows us to capture the effect of distribution at a more granular level. The details are provided in the 'Empirical Context and Data Description' section.

[14] It is possible that customers may make a step-by-step decision on which channel to buy from, which product form to choose, and which brand to buy. A nested structure (such as an NL) can help us understand the potential sequence of such household decision processes. We also test our proposed NL specification against two different NLs, and a multinomial logit (MNL) specification based on the data fit. We find that alternative NLs and MNL specifications can be rejected using the Bayesian Information Criteria.

channels) but also over time (to a lesser extent). Based on our robustness checks, we first find that ignoring the effect of distribution on demand estimation significantly biases the household preference parameters (intrinsic preferences, price disutility, etc.). Further, the use of the price-only demand model as an input in the supply-side equilibrium calculations leads to significantly larger profit margin inferences (up to 55%) compared to equilibrium calculations using the proposed demand model with retail distribution. Second, we find that the observed price and distribution patterns in the data support competition rather than tacit collusion among the studied emerging market manufacturers.

Based on the estimation results, we use a counterfactual study to understand the effect of decreasing distribution costs (for example, due to the construction of new highways and roads by the government) on manufacturers' distribution and price decisions. We find that, in equilibrium, even though firms keep their prices roughly the same, they change their distribution levels significantly as the distribution costs decrease. Specifically, we find that, as distribution costs decrease, 1) both firms increase their distribution levels in the paan-plus channel, and 2) Firm 1 (Firm 2) increases (decreases) its presence in the general stores channel. Further, we find that the aggregate welfares of all three parties – consumers, Firm 1, and Firm 2 – increase as the distribution costs decrease.

In a second counterfactual study, we test the role of exclusive distribution on manufacturers' price and distribution decisions. In a series of simulation studies, we allow 0% (current setting) to 20% of the market to be exclusively covered by the more preferred Firm 1 while allowing the remaining market to be served by both firms together. We find that, in equilibrium, exclusive distribution enables Firm 1 to charge higher prices whereas Firm 2 charges roughly the same prices for both solid and liquid product forms. Regarding the distribution decisions, we find that both firms decrease (increase) their presence in the paan-plus (general) store channel. Regarding the welfares, we find that Firm 1 (Firm 2 and the customers) benefits (hurts) from its (Firm 1's) exclusive distribution. Our counterfactual studies illustrate that asymmetric brand, product form, and channel preferences along with asymmetries in the estimated distribution costs across channels can lead to significant differences among firms in terms of their resource allocation decisions as some characteristics of the marketplace change.

The rest of the paper is organized in the following manner. In the next section, we review the pertinent literature. In Section 3, we discuss our empirical context and data. Section 4 presents our modeling framework and the associated estimation procedure. Section 5 provides our estimation results. Section 6 discusses our managerial implications through our counterfactual studies. Finally, we summarize our findings and contributions, and conclude with caveats and directions for future research.

Literature Review

This study falls in the intersection of two major streams of marketing literature: distribution strategy and marketing in emerging markets. We proceed by discussing recent developments in these two streams of literature along with the research gaps that are filled by this study.

Distribution Strategy

The extant marketing literature has investigated distribution strategies from two perspectives. In the first stream of literature, studies investigate the interactions among different channel members (such as manufacturers, retailers, distributors, etc.). Specifically, these studies have explored conflicts among different channel members (Samaha et al. 2011; Rangan and Jaikumar 1991); coordination and vertical integration of the distribution channel to align the incentives of different channel members (McGuire and Staelin 1983; Desiraju and Moorthy 1997;

Gerstner and Hess 1995; Frazier 1999; Krafft et al. 2015; Anderson and Weitz 1992; Tsay and Agrawal 2004); manufacturers' delegation of power to other channel members and its impact on channel performance (Coughlan and Wernerfelt 1989); negotiations among channel members (in terms of sharing the channel profit) and their impact on channel players' long-term relationships (Srivastava et al. 2000); and the effect of relational history on channel members' performance (Geyskens et al. 1999). We note that our study does not fall into this stream of literature since we do not model manufacturers' relationships with the other channel members (i.e., paan-plus and general stores). This is because 1) the retailers in our setting are small and have limited power compared to manufacturers, and 2) data regarding the interactions among manufacturers and the retail stores, and sales at the retail store level are not available in our context.

The second stream of the distribution strategy literature has explored how manufacturers distribute their products across multiple retail channels and the effectiveness of these channels for manufacturer profitability (for example, Balasubramanian 1998; Cai et al. 2009; Chiang et al. 2003; Neslin et al. 2006; Sharma and Mehrotra 2007; Wallace et al. 2004; Yan 2011). Specifically, studies have investigated the effectiveness of different retail distribution channels (Trivedi 1998; Gonzalez-Benito et al. 2005); their profitability impact for the manufacturer (Padmanabhan and Png 1997); and the effect of product alignment (i.e., which products to sell through) across retail channels (Kumar et al. 2015) and markets (Brynjolfsson et al. 2009) on firm performance.

In this second stream, there are also a few studies that link multi-channel distribution intensity to demand and manufacturer profitability. Yan (2011) investigates the strategic roles of a manufacturer and its retailers in differentiated branding and profit sharing in a multi-channel supply chain context and identifies optimal marketing strategies. Cai et al. (2009) evaluate the impact of different pricing schemes of a manufacturer (selling through dual-channels) on its profitability. They find that consistent (i.e., not varying much over time) pricing schemes can reduce the channel conflict by increasing retailer profits. Sharma and Mehrotra (2007), in the context of a multi-channel environment, show how to choose the optimal channel mix and examine its impact on profitability for a software firm. Pauwels and Neslin (2015) assess the revenue impact of adding brick-and-mortar stores to a firm's already existing repertoire of catalog and internet channels. They show that by adding the physical store channel, the firm in their study can increase its revenue by 20%. Cao and Li (2015) find that cross-channel integration stimulates retailers' sales growth. Jill Avery et al. (2012) investigate whether there is cannibalization and synergy among multiple channels operated by a given firm and, if so, whether such presence affects the demand. They find that the presence of a retail store channel decreases sales only in the catalog channel in the short-run but increases sales in both the catalog and internet channels in the long-run. Our research falls into this second stream of the distribution strategy literature, as we investigate the effectiveness of multi-channel retail distribution strategies for manufacturers; however, we differ from the existing studies since we consider the competition among manufacturers and its impact on their distribution decisions and profitability.

Marketing in Emerging Markets

As emerging markets rapidly grow, multinational companies (MNCs) are becoming more visible in these markets. These MNCs face many challenges (such as an unorganized retail landscape, an insufficient number of stores, etc.) to being competitive in such markets (Dawar and Frost 1999). Although the marketing literature provides abundant guidance on competitive marketing strategies, empirical understanding of competitive strategies (especially in distribution and pricing) in emerging markets is rare.

In terms of distribution, most emerging market firms operate through multiple retail channels, each of which has a different target audience, spread, size, and ownership. The literature documents that retail distribution over multiple channels plays an important role in affecting firm performance in emerging markets (Homburg et al. 2014). For example, Kumar et al. (2015) discuss how an emerging market firm can improve its performance by leveraging its distribution across multiple retail channels. However, there is little research modeling emerging market firms' competition in distribution even at the firm level. Because firms sell their products (in multiple product forms) through different retail channels (with distinct characteristics such as cost structure and attractiveness for customers), implementing competitive distribution strategies at the brand, product form, and retail channel levels can be quite beneficial for increasing firm profitability in emerging markets.

When it comes to pricing, there are limited studies considering the impact of price on sales and firm performance in emerging markets. Since, in most emerging markets, there exist multiple manufacturers selling products within a product category, the effectiveness of their pricing (in terms of profitability) depends on their competitors' reactions. However, rather than considering such competitive pricing, researchers typically rely on conceptual and anecdotal evidence to understand the impact of pricing on firm performance (Arnold and Quelch 1998). Furthermore, as emerging economies continue to grow, customers start to develop needs for a variety of product forms (with similar functionalities). Thus, we extend the existing literature by modeling the emerging market firms' oligopolistic competition in pricing at both the brand and product form levels.

To understand emerging market firms' retail distribution competition along with price, and to guide firms in developing optimal distribution and price levels as the market characteristics change, we build on the existing literature on distribution strategies related to manufacturers' multi-channel distribution and marketing in emerging markets, and also model manufacturers' competition (at the brand, product form, and retail channel levels). See Table 1 for a snapshot of the prior literature and our unique contributions to the marketing literature.

[Insert Table 1 about here]

Empirical Context and Data Description

Our emerging market context is India, where firms rely largely on retail distribution and pricing not only to maximize profit and reach larger target markets but also to respond to competitors' strategies[15]. The Indian retail market is projected to reach $865 billion by 2023 (compared to $490 billion in 2017)[16]. Further, the customer packaged goods (CPG) or fast-moving customer goods (FMCG) market is predicted to grow by 10-12% in the next 10 years. Moreover, regarding retail distribution channels, there has been a significant retail distribution expansion in the last decade. Specifically, the total number of distribution outlets in India is estimated at over 12 million across different industries.

We study the Indian insecticide market, in which manufacturers manage the distribution of their products by managing multiple retail channels. Manufacturers rely on retail audits and past product shipments to evaluate the effectiveness of different retail channels. In this market, it is common for manufacturers which give credits to their retailers to push their products (i.e., manufacturers supply products to retailers with an understanding that the retailers will pay them back after a billing cycle). Manufacturers also incentivize retailers to push their products to end-consumers through recommendations by the retailers' sales personnel. Further, manufacturers manage retailers by offering incentives for generated sales (the cap of sales vary from retailer to

[15] https://www.export.gov/article?id=India-Distribution-and-Sales-Channels.
[16] https://www.export.gov/article?id=India-Distribution-and-Sales-Channels

retailer). Manufacturers also use targeting strategies to reach consumers by targeting different customers with different retail channels. For example, manufacturers target rural and urban customers differently, e.g., paan plus stores are typically used to target both urban and rural customers, whereas general stores are used to target largely urban and semi-urban customers. Moreover, manufacturers use competitive strategies whereby they not only distribute through channels used by competitors but also provide incentives for the retailers in specific competing channels to sell their products.

Our data is provided by one of the manufacturers operating in the insecticide market. The data is observed at the monthly level for a period of 4 years and aggregated to the national level by the data-providing firm[17]. In the data, brand sales at the manufacturer shipping level (in number of units) are observed for two product forms (solid – labeled as Product Form 1, and liquid – labeled as Product Form 2[18]) across two different retail channels (paan-plus – labeled as Channel 1, and general stores – labeled as Channel 2[19]). The data do not provide any evidence of exclusivity in the channel management by the manufacturers. However, such exclusivity is feasible. In the insecticide market, the two major manufacturers capture more than a 70% market share. The rest of the market is composed of many small firms. Because of this, we focus our attention on these two manufacturers and do not endogenize the price and distribution decisions of other small firms.

To understand marketing interactions among competing firms, we obtain firms' monthly pricing information (at the product form level; the unit of price is paisa[20]) and the number of retail

[17] We observe two outlier observations in the data where marketing mix variables differ significantly from the average observations. The data-providing firm is unable to provide any explanation for these observations. To make sure that our results are not affected by these outliers, we conducted a robustness check by removing these two data points and re-estimating our demand model. The model without outliers yield estimates that are very close in magnitude to the model with them. Therefore, we use the model with outliers as our main model for the rest of the analysis (see Table A in Web Appendix A for the details of this robustness check and the comparison of estimates from the two demand models).
[18] The two observed product forms have the same functionality, but different physical forms, e.g. the solid and liquid form of insecticides.
[19] The difference between paan-plus and general stores is the following: paan-plus stores are similar to mom-and-pop stores, whereas general stores are equivalent to small-size retail outlets in the US and Europe markets.
[20] One paisa is equal to one-hundredth of the Indian rupee that is the official currency of India.

stores (at the product form level in each retail channel). Furthermore, we acquire information about the time-variant expected average population size that is served by each channel. This measure is used to weight the total number of stores in each channel over time to construct the unique *distribution* variable that enables us to capture the effect of distribution across different channels at a more granular level. Further, given the lack of data at the regional level, our operationalization enables us to capture some cross-sectional variation in the distribution measure. We observe that on average, Firm 1 (Firm 2) distributes through 64,718 and 420,546 (57,152 and 405,927) weighted paan-plus and general stores, respectively.

We provide descriptive statistics of the key variables in Table 2. As shown in Table 2, on average, Firm 1 charges 4 and 19 paisa higher than Firm 2 in solid and liquid product form, respectively. In terms of the distribution, on average, 1) Firm 1 has more stores than Firm 2 (901 versus 838), and 2) Firm 1 (Firm 2) has 1.26 (1.34) times as many stores in the general stores channel as in the paan-plus stores channel. In terms of sales, on average, Firm 1 has almost the same amount of sales across the two channels, whereas Firm 2 sells 27% more units in general stores than in paan-plus stores. Finally, the average expected population size served by general stores is around 5 times larger than their paan-plus counterparts. All these observed asymmetries across firms, product forms, and retail channels in the key variables motivate us to develop our supply-side competition model to guide firms toward better understanding the existing dynamics in the emerging marketplace.

[Insert Table 2 about here]

Modeling Framework

As discussed earlier, we employ a two-step estimation approach (similar to Che et al. 2007, Sudhir 2001a, and Cosguner et al. 2018). In the first step, we develop (and estimate): 1) an *aggregate market share model* at the brand, product form, and retail channel levels (by accounting for potential endogeneity in price and distribution variables and controlling unobserved consumer heterogeneity); and 2) a *potential market size model* as a function of macroeconomic factors in the relevant emerging marketplace. In the second step, we develop a *multi-product form, multi-channel distribution and price competition model* that takes the estimated *aggregate market share* and *potential market size models* (from the first step) as inputs[21]. We discuss our first- and second-step models along with their associated estimation procedures next.

An Aggregate Market Share Model at Brand, Product Form, and Retail Channel Levels

To develop an econometric brand choice model (at the product form and retail channel levels) with the no-purchase (i.e., outside) option for the typical household h $(h=1,\ldots, H)$, we use a nested logit (NL) specification. In our specification, $k=1, 2$ represents product forms (*1* for solid product form and *2* for liquid product form), $c=1, 2$ represents retail channels (*1* for paan-plus stores and *2* for general stores), and $j=1, 2$ represents manufacturers (firms, products, and brands are used interchangeably) under consideration. The NL tree for the household h's choice at time t is given as follows: at the top choice level (labeled channels), the household chooses which channel to visit $c=1, 2$, or not to visit any channel (i.e., no purchase, $c=0$). At the middle choice level (labeled product forms), the household chooses which product form to choose $k=1, 2$ (given the channel choice from the top choice level). Finally, at the bottom choice level (labeled brands), the household chooses which brand to purchase $j=1, 2$ (conditional on channel and product form

[21] The advantage of the two-step approach is that demand-side estimates become unbiased by any potential misspecification of the game played in the supply side (such as Bertrand, collusion, etc.) since the chosen supply-side specification imposes restrictions in the demand-side estimation as well. In addition, if one intends to test different supply-side games (such as competition versus collusion as we discuss later in the 'Estimation Results' section) to identify the game played in the observed data, the two-step estimation procedure becomes a natural choice, since one can fix the demand-side model first, and then make the supply-side model comparison next.

choices from the top and middle choice levels)[22]. See Figure 1 for the visual representation of the household h's decision tree.

[Insert Figure 1 about here]

Note that, given the NL tree from Figure 1, the household h can either choose to buy brand j in product form k of channel c or the no-purchase option[23] in period t[24]. The choice probability for brand j in product form k of channel c $(j|k,c)$ for household h at time t is as follows:

$$Pr^{ht}(j|k,c) = \frac{\exp\{V_{j|k,c}^{ht}\}}{\sum_{m=1}^{2} \exp\{V_{m|k,c}^{ht}\}}, \quad (1)$$

where $V_{j|k,c}^{ht}$ is the deterministic indirect utility of alternative $j|k,c$ for the household h at time t that is defined as

$$V_{j|k,c}^{ht} = \alpha_{j|k,c}^{h} + \theta X_t{}' + \beta_p^{jht} p_{jt|k} + \beta_d^{h} d_{jt|k,c} + \xi_{jt|k,c}, \quad (2)$$

where $\alpha_{j|k,c}^{h}$ is the intrinsic preference of household h for brand alternative $j|k,c$; X_t is a vector containing seasonality, monthly minimum/maximum temperatures, and monthly average rainfall amount; θ is the corresponding vector of parameters[25]; β_p^{jht} is the disutility of the household h for price of brand $j|k,c$ at time t (i.e., $p_{jt|k}$); β_d^{h} is the utility of the household h from the ease of transportation due to higher availability of the product (e.g., the higher the number of

[22] Our selection of the proposed NL structure is based on both theoretical and empirical robustness. Theoretically, since emerging market consumers suffer the most from accessibility to stores, the first step in a consumer's decision process should be the retail channel choice. Given the channel choice, next, the consumer should decide on the desired product form (solid vs. liquid) since different product forms require different usage needs and infrastructure (for example, the liquid product form needs an electric outlet to be used), and the consumer should decide on her usage needs prior to her brand choice. Finally, given channel and product form choices, the consumer may select her preferred brand. We also communicate with managers of the data-providing (insecticide) firm and confirm that this is indeed the typical decision process for most consumers in the studied insecticide market. To empirically check whether the proposed NL tree is superior, we further estimate two alternative NL demand models: a) product form choice comes first, channel choice comes next, and brand choice comes last; b) brand choice comes first, product form choice comes next, and channel choice comes last. Empirically, the proposed NL model turns out to be superior (BIC=40,090.2M) to these alternatives (BIC=40,090.9M M and BIC=40,091.1M) respectively, based on the Bayesian Information Criteria. The details are available upon request.

[23] We model outside option as the market share of remaining small firms excluded from our specification.

[24] Note that our specification here doesn't allow households to choose multiple product forms at a time. This assumption is consistent with our context since the product forms we study are closely substitutable with each other, i.e., the multiple-discreteness is not prominent in our setting.

[25] Monthly temperature and rainfall information is acquired from the data-providing firm for our observation window. Note that since we study the insecticide market, the number of insects, as well as customers' demand for insecticides, are highly correlated with rain and temperature levels (Wolda 1978). Generally, dry (e.g. low rainfall) and cold weather (e.g., winter season) reduces the insect population; whereas rainy seasons with high temperatures increase it (Porter et al. 1991). Thus, with our utility specification here, in addition to marketing mix variables (i.e., price and distribution), we account for weather-related factors (i.e., seasonality, temperature, and rainfall) that might also impact the demand for insecticides.

stores, the lower the transportation cost, i.e. the higher the utility from the distribution); $d_{jt|k,c}$ is the weighted number of stores (see our discussion in the 'Empirical Context and Data Description' section of how we construct this variable) by which brand alternative $j|k,c$[26] is distributed at time t; and finally, $\xi_{jt|k,c}$ is the demand shock for the brand alternative $j|k,c$ at time t[27].

We model the (brand, household, and time specific) price coefficient β_p^{jht} as follows:

$$\beta_p^{jht} = \beta_p^h + \beta_{pxd}^h d_{jt|k,c}, \qquad (3)$$

where $d_{jt|k,c}$ is the (weighted) number of stores in which brand alternative $j|k,c$ is distributed; β_{pxd}^h is the corresponding response parameter; and β_p^h is the baseline price disutility for household h. Note that this specification of the price coefficient allows the households' price sensitivity for the brand alternative $j|k,c$ to depend on that brand's distribution level. We expect that as the brand alternative $j|k,c$ is sold in more stores, the likelihood of its being sold together with the competing brand increases. This implies the following: the higher the number of stores selling the brand alternative $j|k,c$, the higher the potential price competition for that brand. In other words, we expect the price coefficient to be more negative as $d_{jt|k,c}$ increases. To reiterate, our proposed utility specification captures the effect of distribution in two ways: 1) through the direct effect – a higher level of distribution may increase customers' utility by lowering their transportation burden (i.e., through β_d^h from Equation 2), and 2) through the indirect effect – the level of the focal product's distribution may change the price sensitivity for the focal product (i.e., through β_{pxd}^h)[28].

[26] Inter-channel substitution is not prevalent in our context as the customers' decision to choose a channel is usually governed by their transportation cost, loyalty, and self-selection. We confirm our understanding with managers of the focal firm as well as through customer interviews.

[27] See our 'Price and Retail Distribution Endogeneity' sub-section for how we operationalize the demand shock $\xi_{jt|k,c}$.

[28] It is worth discussing how the baseline price effect (β_p^h), direct distribution effect (β_d^h), and the indirect distribution effect through the interaction of distribution and price (β_{pxd}^h) are empirically identified. The key idea of identification comes from the observed variations in price and distribution variables that yield variations in the observed market shares. We illustrate how the empirical identification is possible in our setting through a stylized example in our Web Appendix B. In the same web appendix, we also provide a microsimulation study to illustrate how we are able to identify the assumed demand parameters (including β_p^h, β_d^h , and β_{pxd}^h) from the simulated data (see our Web Appendix B for the details).

Given the deterministic indirect brand utilities from Equation (2), product form choice probabilities *k=1,2* for household *h* at time *t* becomes:

$$Pr^{ht}(k|c) = \frac{exp\left\{\lambda_{k|c} IV_{k|c}^{ht}\right\}}{\sum_{m=1}^{2} exp\left\{\lambda_{m|c} IV_{m|c}^{ht}\right\}}, \tag{4}$$

where $k|c$ is product form *k* of channel *c*, $IV_{k|c}^{ht} = \ln(\sum_{j=1}^{2} \exp\{V_{j|k,c}^{ht}\})$ is the inclusive value (IV) for product form $k|c$ for household *h* at time *t*, and $\lambda_{k|c}$ are the corresponding IV parameters. Finally, the channel choice probabilities *c=0,1,2* for household *h* at time *t* becomes the following.

$$Pr^{ht}(c) = \frac{\exp\{V_c^{ht}\}}{\sum_{m=0}^{2} \exp\{V_m^{ht}\}}, \tag{5}$$

where $V_c^{ht} = exp\{\lambda_c IV_c^{ht}\}$ for *c=1,2* and $V_0^{ht} = 0$ (i.e., the deterministic indirect utility for the outside good is normalized to zero), $IV_c^{ht} = \ln\left(\sum_{k=1}^{2} exp\left\{\lambda_{k|c} IV_{k|c}^{ht}\right\}\right)$ is the IV for channel *c=1,2* for household *h* at time *t*, and λ_c *(c=1,2)* are the corresponding IV parameters.

Given the household-specific brand, product form, and channel choice probabilities from Equations (1), (4), and (5), the probability of choosing brand *j=1,2* in product form *k=1,2* of channel *c=1,2* by household *h* at time *t* becomes the following:

$$Pr^{ht}(j,k,c) = Pr^{ht}(c) Pr^{ht}(k|c) Pr^{ht}(j|k,c). \tag{6}$$

Given the household-level choice probability from Equation (6), the aggregate market share of brand *j* in product form *k* of channel *c* at time *t* ($MS_{j,k,c,t}$) can be integrated over the household heterogeneity distribution as follows:

$$MS_{j,k,c,t} = \int_h Pr^{ht}(j,k,c)\, \phi(h) dh, \tag{7}$$

where the $\phi(h)$ is the joint density of the unobserved household heterogeneity distribution (see our 'Unobserved Heterogeneity' subsection for details).

Potential Market Size

Since our context is an emerging marketplace, the potential market size (M_t) is expected to change over time. To capture such variation in M_t, we model M_t as a function of observed macroeconomic factors of the relevant emerging marketplace as follows:

$$M_t = W_t \zeta, \tag{8}$$

where, W_t contains India's population, the unemployment rate, and the GDP of India at time *t*, and ζ is the corresponding vector of parameters.

Note that, given the aggregate market share of brand *j* in product form *k* of channel *c* at time *t* (i.e., $MS_{j,k,c,t}$) from Equation (7) and the potential market size at time *t* (i.e., M_t) from Equation (8), the aggregate sales of brand *j* in product form *k* of channel *c* at time *t* ($S_{j,k,c,t}$) can be calculated as follows:

$$S_{j,k,c,t} = MS_{j,k,c,t} * M_t. \tag{9}$$

As discussed earlier, Equation (9), i.e., aggregate sales, becomes an input into our supply-side distribution and price competition model.

Price and Retail Distribution Endogeneity

Firms' other marketing decisions that are unobserved (by the researcher) might be set together with their (product form-level) price and (product form- and channel-level) distribution decisions. Thus, firms' price and distribution decisions might be endogenous (Pattabhiramaiah et al. 2017). To account for the potential endogeneity of price and distribution variables, we use a control function approach (Petrin and Train 2010, an idea similar to the approach in Villas-Boas and Winer 1999). This approach involves running first-stage linear regression models of price ($p_{jt|k}$) and distribution ($d_{jt|kc}$) on instruments.

Due to difficulties in finding valid (and strong) instruments to account for the potential endogeneity problem in the price and distribution variables, we review existing marketing studies to identify instruments used in the literature (Ataman et al. 2010; Kumar et al. 2015; Pancras and Sudhir 2007). Based on this review, we find that the widely used instruments (in the literature) to account for price (retail distribution) endogeneity are 1) the pricing (distribution) levels of firms in similar markets; 2) the cost of raw materials (diesel/gasoline); and 3) past performance metrics such as differences in lagged sales. Note that since our setting is the entire Indian market, it is not feasible for us to acquire marketing instruments from other similar markets (with similar retail channels, customer/firm characteristics, and behaviors). Regarding the other potential instruments discussed, we use a combination of instruments to account for the potential price and distribution endogeneity problem. First, we collect *time-variant prices* of *raw materials* – $Cost\_Raw_{j,t}$ for firm j at time t used in the production of insecticides – and use these as instruments to account for price endogeneity. The use of raw material costs to control price endogeneity is quite common in the literature (see, for example, Cosguner et al. 2018; Pancras and Sudhir 2007). Raw material costs should impact the pricing decision of a firm's product. However, such costs are unlikely to impact the specific sales of the firm, at least directly. Following similar logic, we use the cost of diesel – $Diesel_t$ at time t as an instrument to account for the retail distribution endogeneity problem. Second, managers might look at their firm's earlier performance and make their current marketing mix decisions accordingly. Along this line, we use *changes in sales from time t-2 to t-1* (denoted by $\Delta Sales_{jk,t-2\rightarrow t-1}$) as an additional instrument to account for potential price and distribution endogeneity problems. Changes in the sales should impact the efficacy of the marketing mix at t; however, it should not impact the demand itself at t. The choice of the sales difference as an instrument is also consistent with the existing marketing studies (Ataman et al. 2010). In the end, our first stage regression equations become:

$$p_{jt|k} = \delta_1 Cost\_Raw_{j,t} + \delta_2 \Delta Sales_{jk,t-2\rightarrow t-1} + w_{jt|k} \tag{10a}$$

$$d_{jt|k,c} = \delta_3 Diesel_t + \delta_4 \Delta Sales_{j,k,c,t-2\rightarrow t-1} + w_{jt|k,c}. \tag{10b}$$

First-stage regressions yield F-statistics that are significantly larger than 10. In addition, we obtain R^2 measures of 55.3% and 42.1% (on average) from Equations (10a) and (10b), respectively. We check the validity of our instruments using a correlation analysis and conduct a modified Sargan test. These analyses show that our instruments are both valid and strong[29].

We label $\widehat{w_{jt|k}}$ as the fitted firm- and product form-specific price residual, and $\widehat{w_{jt|k,c}}$ as the fitted firm-, product form-, and channel-specific distribution residual for firm j for each month. We use linear functions of these residuals, $\varphi_{j|k}\widehat{w_{jt|k}}$ (for the price) and $\varphi_{j|k,c}\widehat{w_{jt|k,c}}$ (for distribution), to approximate the demand shock $\xi_{jt|k,c}$ in Equation (2). The assumption of the control function approach is that, conditional on $\widehat{w_{jt|k}}$ $\widehat{w_{jt|k,c}}$, the Type I Extreme Value error term $\varepsilon_{jt|k,c}$ (from the bottom choice level of the NL, i.e., brand choice) becomes independent from $p_{jt|k}$ and $d_{jt|k,c}$. This approach has been used widely in the literature to control for the potential endogeneity problem in marketing variables (see, for example, Ma et al. 2005; Zhang et al. 2017).

Unobserved Heterogeneity

Different households may have different intrinsic preferences for the different firm, product form, and retail channel combinations, and further, they may respond to marketing mix variables differently. To control for such unobserved household-level heterogeneity, we use the random coefficient specification (Keane and Wasi 2013; Park and Gupta 2009). The use of the random coefficient specification in the estimation of choice models with aggregate sales data (such as ours) is very common in the literature (see, for example, Chintagunta 2001; Sudhir 2001b).

[29] The correlation between the dependent variable and exclusion restrictions ranges between 0.12 and 0.21 suggesting that our instruments are valid. The modified Sargan test for over-identification of instruments further supports the validity of our instruments. In addition, the correlation between endogenous variables and instruments ranges between 0.38 and 0.79, suggesting that the instruments are strong. We confirm our intuition regarding the instruments with managers of the data-providing firm. Managers state that they consider both cost-related instruments and lagged sales differences to decide on their price and distribution strategies at time *t*. We have also tested for the potential serial correlation in demand and did not find any evidence of the same.

Specifically, we assume that the household-level preference parameters in Equations (2) and (3) come from the following distributions: $\alpha_{j|k,c}^{h} \sim N(\mu_{\alpha_{j|k,c}^{h}}, \sigma_{\alpha_{j|k,c}^{h}}^{2}), \beta_{p}^{h} \sim N(\mu_{\beta_{p}^{h}}, \sigma_{\beta_{p}^{h}}^{2})$ and $\beta_{d}^{h} \sim N(\mu_{\beta_{d}^{h}}, \sigma_{\beta_{d}^{h}}^{2})$, where $\mu_{\alpha_{j|k,c}^{h}}$, $\mu_{\beta_{p}^{h}}$ and $\mu_{\beta_{d}^{h}}$ ($\sigma_{\alpha_{j|k,c}^{h}}^{2}$, $\sigma_{\beta_{p}^{h}}^{2}$ and $\sigma_{\beta_{d}^{h}}^{2}$) are mean values (variances) of households' intrinsic preference for firm *j* in product form *k* of channel *c*, disutility for price, and utility for distribution (for ease of access to products), respectively.

Supply-Side Price and Distribution Competition Model

For our empirical application, as noted before, we focus on two major insecticide firms (i.e., *J=2*), two major product forms (liquid and solid, i.e., *K=2*), and two major retail channels (paan-plus and general stores, i.e., *C=2*). We define the profit function for the j^{th} *(j=1,2)* firm at time *t* as follows:

$$\pi_{jt} = \sum_{c=1}^{2} \sum_{k=1}^{2} \left(p_{jt|k} - mc_{jt|k}\right) S_{j,k,c,t} - \sum_{k=1}^{2} \sum_{c=1}^{2} dc_{jt|k,c} * {d_{jt|k,c}}^{2}, \tag{11}$$

where $mc_{jt|k}$ is the time-variant (marginal) production cost for brand *j* in product form *k* at time *t*, $dc_{jt|k,c}$ is the time variant parameter of the convex[30] distribution cost function (Gallego and Wang 2014; Ghosh and Shah 2015) of firm *j* in product form *k* of channel *c* at time *t*[31], and $S_{j,k,c,t}$ is the aggregate demand function from Equation (9).

[30] We make the convex cost assumption for two reasons. First, the convexity assumption is needed to keep supply-side equilibrium calculations (for the counterfactual studies) computationally tractable. Second, due to the abundance of underdeveloped rural markets in India, as confirmed by the managers of the data-providing firm, increasing the number of stores creates sufficiently large costs for firms. For example, if a firm starts distributing in various far-reach areas to fulfill the demand, the firm needs to buy new trucks, hire more drivers, and obviously pay more for the diesel. Therefore, in the studied marketplace, increasing distribution levels magnifies the cost of distribution exponentially. As a robustness check, we estimate the distribution cost under the linear cost assumption. Since magnitudes of convex and linear distribution costs are not directly comparable, we only report the results with the convex cost here. However, the results with the linear distribution cost are also available upon request.

[31] We acknowledge that our retail distribution cost specification does not capture the fixed costs of initializing new retail distribution agreements between a firm and its stores (such as the cost of initial negotiations between the firm and store managers). Instead, our specification captures the costs of maintaining retail distribution relationships between the firm and its stores that have already agreed to distribute the firm's products. Modeling such fixed costs is not empirically possible with our current dataset since we are not able to distinguish between a firm's new stores and its repeat-distributing stores; instead, we just observe the total number of stores distributing the firm's products. Even though modeling such fixed costs is not possible in our current setting, we believe that such fixed costs are reasonably small in the studied marketplace due to the small sizes of stores compared to typical grocery chains in most developed markets. Thus, it is relatively easier (i.e., less costly) for the studied firms to get stores to agree to sell their products in our emerging market context. However, if such distribution information (previously versus recently acquired stores) is available, we believe that modeling such fixed costs is an important, but challenging area of future research.

Under the Bertrand pricing and distribution assumption for the game played among manufacturers, and the profit function of firm *j* at time *t* from Equation (11), the first-order conditions for the *j*[th] firm's profit with respect to its product form-level prices can be written as:

$$\frac{d\pi_{jt}}{dp_{jt|k}} = S_{j,k,t} + \sum_{k=1}^{2}\left(p_{jt|k} - mc_{jt|k}\right)\frac{\partial S_{j,k,t}}{\partial p_{jt|k}},\ k=1,\ 2, \tag{12}$$

where $S_{j,k,t} = \sum_{c=1}^{2} S_{j,k,c,t}$ and $\frac{\partial S_{j,k,t}}{\partial p_{jt|k}} = \sum_{c=1}^{2}\frac{\partial S_{j,k,c,t}}{\partial p_{jt|k}}$. Note that this summation across channels is possible since firms have identical profit margins across different channels, i.e., pricing decisions are not made at the channel level[32]. Setting these first-order conditions in Equation (12) to zero at the firm and product form levels, and solving them simultaneously, the time-variant cost of production ($mc_{jt|k}$) can be inverted as follows[33]:

$$\widehat{mc_{jt|k}} = p_{jt|k} + \left[S_{j,-k,t} * \frac{\partial S_{j,-k,t}}{\partial p_{jt|k}} - S_{j,k,t} * \frac{\partial S_{j,-k,t}}{\partial p_{jt|-k}}\right] / \left[\frac{\partial S_{j,1,t}}{\partial p_{jt|2}} * \frac{\partial S_{j,2,t}}{\partial p_{jt|1}} - \frac{\partial S_{j,1,t}}{\partial p_{jt|1}} * \frac{\partial S_{j,2,t}}{\partial p_{jt|2}}\right] \tag{13}$$

To invert time-variant distribution costs, we plug the inverted time-variant marginal costs from Equation (13) into the profit function from Equation (11), and then calculate the first-order conditions for firm *j*'s profit with respect to product form *k* of channel *c* distribution as follows:

$$\frac{\partial \pi_{j,t}}{\partial d_{jt|k,c}} = \sum_{m=1}^{2}\sum_{c=1}^{2}\left(p_{jt|m} - \widehat{mc_{jt|m}}\right)\frac{\partial S_{j,m,c,t}}{\partial d_{jt|k,c}} - 2dc_{jt|k,c}d_{jt|k,c},\ c=1,2,\ k=1,2. \tag{14}$$

By setting the first-order conditions above to zero, we can invert the time-variant distribution cost ($dc_{jt|k,c}$) as the following:

$$\widehat{dc_{jt|k,c}} = \frac{\sum_{m=1}^{2}\sum_{l=1}^{2}\left(p_{jt|m} - \widehat{mc_{jt|m}}\right)\frac{\partial S_{j,m,l,t}}{\partial d_{jt|k,c}}}{2d_{jt|k,c}}. \tag{15}$$

[32] The pricing decision at the retail channel level is computationally feasible if firms charge different prices across different channels. However, that is not the case in our study context since we don't observe any price variations across different channels for brands in the same product form.

[33] Note that, based on Equation (13), $\widehat{mc_{jt|k}}$ gets smaller as i) the demand of the focal (i.e., *k*) and the competing (i.e., *-k*) product forms ($S_{j,k,t}$ and $S_{j,-k,t}$) get larger; ii) the cross price derivatives of the focal and the competing product form's demands ($\partial S_{j,k,t}/\partial p_{jt|-k}$ and $\partial S_{j,-k,t}/\partial p_{jt|k}$) gets larger; iii) the (absolute value of the) own price derivative of the focal product form demand ($|\partial S_{j,k,t}/\partial p_{jt|k}|$) gets smaller; iv) if $S_{j,k,t} > |\partial S_{j,k,t}/\partial p_{jt|k}|$, the (absolute value of the) own price derivative of the competing product form demand ($|\partial S_{j,-k,t}/\partial p_{jt|-k}|$) gets larger.

As seen in Equations (13) and (15), we have closed-form expressions for the production and distribution costs (given the aggregate sales model from Equation (9) and the observed prices and distribution levels in the data). We acknowledge that the identification of the costs in Equations (13) and (15) relies on our supply-side Bertrand competition assumption of the game played among manufacturers. In other words, the inverted costs in Equations (13) and (15) would be different (but still identifiable), if a different game assumption was made in the supply side. Accordingly, we estimate an alternative game in which firms are assumed to be in tacit collusion to check whether our Bertrand assumption is a reasonable one. See the "Do Firms Collude in the Insecticide Market?" subsection under the "Estimation Results" section for the details of that comparison.

Model Estimation

As discussed earlier, we estimate our aggregate market share (at the brand, product form, and retail channel levels) and potential market size models in our first-step estimation. Parameters of the aggregate market share model are estimated by maximizing the log of the simulated sample likelihood below[34]:

$$L = \prod_{t=1}^{T}\{\prod_{j=1}^{J}\prod_{k=1}^{K}\prod_{c=1}^{C} MS_{j,k,c,t}{}^{D_{j,k,c,t}}\} * MS_{0,t}{}^{D_{0,t}}, \tag{16}$$

where $D_{j,k,c,t}$ is the sales of brand j (in product form k of channel c at time t); $D_{0,t}$ is the sales of the outside good; and $MS_{j,k,c,t}$ is the aggregate market share of brand j (in product form k of channel c at time t) defined in Equation (7). Parameters of the potential market size model (i.e., M_t) in Equation (9) are estimated via the ordinary least squares (OLS) method.

[34] We acknowledge that it is more common to use Berry's (1994) formulation of logit (NL in our specification) to estimate demand with market-level sales data. Instead, we implement a simulated likelihood-based approach for the following reason. As discussed by Park and Gupta (2009), Berry (1994) assumes that the observed market shares of alternatives (brands, product forms, and channels in our case) should have no sampling error. In other words, the randomness in market shares only comes from unmeasured product characteristics. Thus, if the sampling error in shares is small, Berry (1994) can provide consistent estimates of the demand-side parameters. Whereas, in our case, there might be sampling errors in market shares since 1) the size of the households is evolving over time due to our emerging market setting; 2) the product category studied is somewhat seasonal, as opposed to a typical repeat purchase category, i.e., the number of customers may be varying over time; and 3) the number of stores carrying the products also evolve over time, i.e., the size of the covered market may change over time. Park and Gupta (2009) show that the simulated likelihood-based estimation method can allow such sampling errors in market shares and still yield unbiased and efficient demand parameter estimates. Therefore, we implement a simulated likelihood-based approach to estimate our demand-side parameters.

The aggregate sales model (i.e., $S_{j,k,c,t}$) defined in Equation (9) is used as input into our second-step estimation that involves the inversion of time-variant production and distribution costs defined in Equations (13) and (15), respectively. Once production (distribution) costs are inverted, we pool these production (distribution) costs across firms and product forms (across firms, product forms, and retail channels). Then, we estimate our production (distribution) cost function by modeling these pooled production (distribution) costs as a function of the dummy variables of each firm and product form combination (each firm, product form, and channel combination) and years, and the interaction of these dummy variables.

Before discussing our estimation results, we would like to note that, in this study, we have not considered retailers' pricing rules (i.e., we assume that retailers are not strategic, acquiring a fixed and small percentage of the entire channel's profit margin[35]) unlike some previous research studying vertical pricing interactions (e.g., Sudhir 2001a; Villas-Boas 2007) in distribution channels. We make this assumption for the following reasons. First, as explained earlier, the retail industry is unorganized in most emerging markets (Jerath et al. 2016). For example, 85% of the retail industry in BRIC countries is unorganized[36]. The retail industry in emerging markets is mostly dominated by small mom-and-pop and grocery channels, which are individually owned, small in size, low in capital investment, but large in number (Sarma 2005). Chain retailing or organized retail businesses are not as prevalent in emerging markets as in developed markets, although there is a slow transition to the adoption of chain retailing (modern stores) (Narayan et al. 2015). Hence, the bargaining power of retailers is very limited in emerging markets. In addition, since the size of emerging markets is large and the retail infrastructure is not well-developed (Sheth

[35] In our application, since the retailers' profit margin percentage is not observed, rather than assuming an arbitrary percentage (such as 5% or 10%) for the retailer, we assume that the entire channel margin goes to manufacturers. If the retailers' percentage margins are observed, the profit function in Equation (11) can easily be modified, and production and distribution costs from Equations (13) and (15) can be inverted accordingly.
[36] http://rasci.in/downloads/2007/The_BRIC_Promise_2007.pdf

2011), receiving supply from manufacturers regularly is more important (to satisfy customer demand) than strategically setting prices for small emerging market retailers. Finally, in our empirical setting, it is not feasible for us to model the retailers' pricing roles since we do not have sales data at the individual store level.

Estimation Results

In this section, first, we discuss our demand-side model selection by showing the importance of considering households' sequential choice process (the channel first, product form next, and brand last) and the role of distribution in the demand estimation. Second, we discuss our demand-side estimation results. Third, we discuss our supply-side estimation results. Fourth, we discuss implications of ignoring the role of distribution in estimating firms' profit margins. Finally, we discuss our robustness check regarding the identification of the supply-side game played in the data, i.e., Bertrand competition versus collusion.

Demand Side Model Selection

Our demand-side estimates are reported in Table 3. We estimate three different demand models: 1) multinomial logit demand with price and distribution (labeled as Model 1); 2) nested logit demand without distribution (labeled as Model 2); and 3) nested logit demand with both price and distribution, i.e., our proposed demand model (labeled as Model 3)[37]. As expected, the NL demand model with both price and distribution[38], i.e., Model 3, outperforms both Models 1 and 2 based on the BIC ($BIC_{MODEL\ 1}$=40,092.7M, $BIC_{MODEL\ 2}$=40,113.0M and BIC_{MODEL}

[37] In all three estimated demand models, we control for both the potential endogeneity problem (in price and distribution variables) and unobserved customer heterogeneity. Under our proposed NL structure, if we drop both the endogeneity correction and heterogeneity, the BIC value increases to 40,117.9M from 40,090.2M. If we account for endogeneity without controlling for heterogeneity, the BIC value becomes 40,095.6M. This suggests that, in our emerging market setting, accounting for endogeneity explains relatively more data variations compared to controlling consumer heterogeneity.

[38] Regarding the starting values of the parameters in the estimation, we first estimated the homogenous multinomial logit (MNL) model in which the converged set of parameters globally maximizes the log-likelihood. We next use the estimated parameters from the homogeneous MNL as the starting values of the homogenous NL model. Last, we use the estimated parameters from the homogenous NL model to estimate the parameters of the heterogeneous NL model. To simulate the log-likelihood under the heterogeneous NL case, we use a set of R=500 i.i.d. standard Normal draws that are drawn at the seed=1.

$_3$=40,090.2M).[39]. Due to the inferior data fit of Models 1 and 2, we use Model 3 as our main model for further analysis.

[Insert Table 3 about here]

Demand-Side Estimation Results

As seen in Table 3, based on the estimated mean intrinsic preference parameters, customers prefer Firm 1, the solid product form, and paan-plus stores to Firm 2, the liquid product form, and general stores, respectively. As expected, our estimated mean price and distribution coefficients turn out to be negative (-0.04) and positive (0.003), respectively. This suggests that as the price of a brand-product form (the distribution level of a brand-product form-channel) combination increases, customer utility for the corresponding combination decreases (increases). This finding suggests that there is a direct positive effect of distribution (through lowering the transportation burden) on customer utilities. As discussed before, the level of distribution might also have an indirect effect through changing households' price sensitivities. Our results suggest that as the level of distribution for a brand-product form-channel combination increases, customers' price sensitivity for that combination also increases. This finding highlights the importance of modeling households' price disutility as a function of the corresponding brand-product form-channel combination's distribution level to be able to understand the true pricing responses of households.

Regarding unobserved heterogeneity, we find that households in the insecticide market are highly heterogeneous in terms of their intrinsic preferences and their responsiveness to marketing mix variables. We find estimates for price and distribution endogeneity controls to be significant. Further, our results suggest that seasonality, temperature, and rainfall controls are significant,

[39] Note that the comparison of Model 1 and Model 3 shows that there are significant differences in the sizes of estimated preference parameters if one does not consider households' sequential decision process in the estimated choice model (i.e., use of MNL rather than NL). Further, the comparison of Models 2 and 3 shows that ignoring the role of distribution also causes significant biases in the estimated preference parameters. For example, the price coefficient is 24% underestimated in Model 2 compared to Model 3, suggesting that it is important to consider the effect of distribution in the demand estimation even if one's objective is solely to understand households' responses to prices to make sales forecasting.

suggesting that controlling weather-related factors is also important in understanding households' demand for insecticides. Finally, we find that estimates for inclusive value parameters (of our NL specification) are significant and varying in sizes across different channel and product form nests, suggesting that the nested structure helps us capture meaningful variations in the observed sales data.

Supply-Side Estimation Results

To estimate the production cost function, we first pool the inverted (marginal) production costs from Equation (13) across firms and product forms. Second, we estimate a linear regression model by using the pooled production costs as our dependent variable, and firm-product form dummies[40], year dummies[41], and the interaction of these dummies[42] as our independent variables. The parameter estimates of this regression model are reported in the first column of Table 4 as Model 1. As seen in Table 4, all firm-product form dummies, year dummies, and their interactions are significant, implying that production costs not only differ across different firms and product forms but also over time. Further, the estimated production costs are in the same ballpark as the actual cost estimates of managers of the data-providing firm. The differences in costs between model predictions and managerial estimates vary between -8.73% to +10.12%. That provides face validity to our cost estimates.

[Insert Table 4 about here]

[40] Production costs might differ across different firms and within the same firm across different product forms. That's why we use firm-product form dummies. Further, note that the inverted production costs are nothing but the observed prices minus optimal margins predicted by our supply-side model. In other words, the variations in the inverted production costs are the unexplained variations (by our model) in observed prices. Since pricing levels may differ not only across firms but also within the same firm across product forms, firm-product form dummies can be used to capture such unexplained variation in observed prices that are embedded into the recovered production costs.
[41] Since the macroeconomic environment in our emerging market is not constant, and since firms might be investing in their production technologies through R&D investments, we expect that firms' marginal cost of production may also change over time. Accordingly, we control these time-variant effects through the year dummies in our production cost regression.
[42] It is possible that different firms might invest differently in improving their production technologies (across different product forms) over time. Further, macroeconomic changes in the emerging marketplace might change production cost structures differently for different product forms. To capture such dynamics, we use the interactions of our firm-product form dummies and year dummies as additional controls.

To estimate the distribution cost function, similar to the production cost function estimation, we first pool the inverted distribution costs from Equation (15) across firms, product forms, and retail channels. Next, we estimate a linear regression model by using the pooled distribution costs as our dependent variable, and firm-product form-channel dummies[43], year dummies, and the interaction of these dummies as our independent variables. The parameter estimates of this regression model are reported in the first column of Table 5 as Model 1. As seen in Table 5, the estimates suggest that distribution costs differ not only across different firms, product forms, and retail channels but also over time to a smaller extent.

[Insert Table 5 about here]

Note that our proposed supply-side model can easily be used by emerging market firms to understand their competitors' cost asymmetries. This information carries an important value for firms in optimizing their decisions and calculating their competitors' reactions. Moreover, understanding the cost of distribution and production might be useful for firms that are planning to enter into the insecticide market to assess the profitability of their entry decisions.

Role of Retail Distribution in the Supply-Side Estimation

In this subsection, we aim to illustrate the role of ignoring distribution on profit-maximizing firms' pricing decisions. Since the previous models studying firm competition ignore the role of distribution and solely focus on price optimization, to achieve our objective, we estimate a price-only supply-side model and compare that benchmark model to our proposed supply-side model with price and distribution. For the price-only model, we use the estimated price-only demand model (Model 2) in Table 3 as the input and derive the marginal production cost in

[43] Similar to our previous discussion, distribution costs may not only differ across firms but also within the same firm and across different product forms and retail channels. Further, distribution costs can differ over time due to i) changes in the macroeconomic environment; and ii) firms' investments in their transportation technologies and storage/warehouse infrastructures/facilities.

Equation (13). Next, we estimate a production cost function similar to Model 1 in Table 4 with firm-product form dummies, year dummies, and their interactions. We label this model as Model 2 and report its estimated parameters in Table 4. Our proposed model, i.e., Model 1, outperforms the price-only model, i.e., Model 2, both in terms of R^2 (95.97% versus 92.09%) and SSR (1,002.24 versus 1,421.90) criteria. This suggests that ignoring distribution significantly worsens the fit of the estimated production cost function[44].

Next, we use the estimated parameters from Models 1 and 2 (in Table 4) to calculate the optimal profit margins for each firm and product form combination over our observed data span of four years. Results related to the comparison of the calculated equilibrium profit margins are reported in Table 6. As evident from Table 6, ignoring distribution causes substantive over-estimation of optimal profit margins ranging from 7% (Firm 1 in solid-form on Year 2) to 55% (Firm 1 in liquid form on Year 4). These results suggest that the incorporation of distribution along with price in firms' profit maximization objectives is indeed a critical requirement for emerging market firms and ignoring the distribution might cause firms to set prices that are significantly higher than optimal levels.

[Insert Table 6 about here]

Do Firms Collude in the Insecticide Market?

Note that we assume firms (in our emerging market context) compete in both price and distribution. However, there is a possibility that firms may tacitly collude (rather than compete) and decide on price and distribution levels based on maximizing the combined profits in the

[44] Note that inverted marginal production costs under both the proposed and price-only models correspond to the unexplained parts of the observed price variations by the proposed and price-only models, respectively. Thus, comparing SSR values under both the proposed and price-only scenarios tells us which supply-side specification explains the observed price variations better, i.e., the model with smaller SSR value explains the observed pricing patterns better. A similar idea has been implemented by Cosguner et al. (2018) to compare supply-side pricing models (dynamic versus myopic in their setting).

insecticide category[45]. To show that our assumption of Bertrand competition in pricing is consistent with our data, we estimate a model of collusive pricing. See Model 3 in Table 4 for the estimated production cost function under this collusive pricing assumption. Results suggest that our proposed production cost model (Model 1 in Table 4) performs significantly better (R^2=95.97% and SSR=1,002.4) than the production cost model under the collusive pricing assumption (R^2=87.78% and SSR=7,026.77). This suggests that the observed price variations in the data can be better explained by the proposed price competition model than the alternative collusive pricing model.

Even though firms compete in prices, there is still a possibility that they might be colluding in their distribution decisions. To test whether the observed data supports collusion (rather than competition), we estimate the distribution cost function under the collusion assumption (see Model 2 in Table 5 for the estimates of this alternative model). Similar to the collusive pricing model, we find that the distribution cost model under the collusive distribution assumption has a poor fit in comparison to our proposed model (R^2 of 96.46% versus 98.86%; and SSR of 4,822.5M versus 2,752.9M). Based on the inferior fit performance of the (price and distribution cost) models under the tacit collusion assumption and the respective performances of our proposed competitive models, we conclude that our emerging market firms do not collude in the studied insecticide market.

Managerial Implications

In this section, based on the estimation results, we conduct a series of counterfactual studies to examine the policy implications of our distribution and price competition model. Specifically, our objective is to investigate what happens to prices, distribution levels, profits of firms, and

[45] We thank the anonymous reviewer for pointing out this important point.

consumer surplus (in equilibrium) as 1) firms' distribution costs decrease and 2) one firm starts to distribute exclusively.

Decreasing Distribution Costs

Since our context is an emerging market, it is possible that the government may improve the transportation infrastructure in India by constructing new roads and highways, and thus it may become easier and ultimately less costly for product manufacturers to distribute their products in the future. Firms may respond to such decreases in distribution costs by changing their price and distribution levels asymmetrically across different product forms and retail channels. Therefore, our objective is to understand such asymmetries among firms in responding to decreases in distribution costs. To achieve this objective, we gradually decrease the estimated distribution costs from 0% to 20% by 0.1% increments (by keeping all else – marginal cost and demand-side estimates – constant) and numerically solve for the equilibrium of the supply-side (distribution and price) competition game for each candidate set of distribution costs. We plot how equilibrium prices, distribution levels, profits, and consumer surplus change as the distribution costs decrease in Figures 2A, 2B, 2C, and 2D, respectively.

[Insert Figure 2 about here]

As seen in Figures 2A and 2B, firms respond to decreases in the distribution costs by changing their distribution levels rather than prices. Figure 2B illustrates that both firms respond to decreases in distribution costs by increasing their distribution level in the paan-plus stores. This is because paan-plus stores are more preferred by customers than general stores (based on our demand-side estimation results). In other words, increasing presence in paan-plus stores brings a higher sales return (on average) for both firms compared to general stores. Due to this relatively higher sales return, both firms increase their presence in paan-plus stores as the distribution costs decrease. Unlike paan-plus stores, in general stores, firms respond to decreases in the distributions costs asymmetrically: Firm1 increases its distribution level, whereas Firm 2 does the opposite. This happens since Firm 1 is significantly more preferred compared to Firm 2 in that channel (based on our demand-side estimation results). In other words, even though general stores are less preferred than paan-plus stores, due to the decrease in the distribution cost in this channel, Firm 1 prefers to increase its presence in general stores since the sales return of this investment is still significant (due to Firm 2's inferior brand preference). Whereas Firm 2, because the sales return from investing in the general stores channel is only marginal (due to its inferior brand preference), prefers to decrease its presence in this channel and shift its distribution resources into the paan-plus channel. This strategy helps Firm 2 to be more competitive against the more preferred Firm 1 in the paan-plus channel. Regarding the equilibrium profits and consumer surplus, as seen in Figures 2C and 2D, the welfare of all three parties – Firm 1, Firm 2, and consumers – increases as the distribution costs decrease.

Exclusive Distribution of Firm 1

Although the two manufacturers in the insecticide market do not distribute exclusively (as evident from the available data), to be able to mitigate the competition, they may consider exclusively distributing their products in some parts of the emerging marketplace. Such exclusivity may help firms not only to charge higher prices but also to effectively reallocate their distribution resources to improve their profits. To understand the implications of such exclusivity on policies, profits, and consumer surplus, we allow the more preferred Firm 1 to cover 0% to 20% of the market exclusively with 0.1% increments (by keeping all else – production and distributions costs and demand-side estimates – constant), and numerically solve for the equilibrium of the supply-side (distribution and price) competition game for each exclusive coverage percentage (0% to

20%) of Firm 1. We plot how equilibrium prices, distribution levels, profits, and consumer surplus change as the exclusivity percentage of Firm 1 increases in Figures 3A, 3B, 3C, and 3D, respectively.

[Insert Figure 3 about here]

As seen in Figure 3A, Firm 1 increases its prices in both solid and liquid product forms as its exclusive coverage increases because the exclusivity helps Firm 1 to avoid competition with Firm 2 in the corresponding exclusive segment. Regarding Firm 2's pricing, as seen in Figure 3A, Firm 2 can keep its prices relatively the same as the exclusivity of Firm 1 increases (or, the potential market size for Firm 2 decreases). Figure 3B shows that both Firm 1 and 2 increase (decrease) their distribution levels in general (paan-plus) stores. Regarding Firm 1, this happens for two reasons: 1) because of exclusivity (i.e., the lack of the competitor's products in the exclusive segment), the less preferred general stores become more attractive in terms of bringing sales for Firm 1; and 2) moving distribution resources from paan-plus to general stores saves costs since distributing to the former channel is costlier (based on the supply-side estimation results) than the latter. Regarding Firm 2, given that the potential market size decreases due to Firm 1's exclusive coverage, sales return from distribution decreases for Firm 2. Thus, Firm 2 shifts its distribution from paan-plus to general stores to cut its distribution costs, since it is less costly to distribute to general stores (based on the supply-side estimation results). Finally, regarding the profits and consumer surplus, Figures 3C and 3D show that the exclusivity of Firm 1 increases the welfare of Firm 1 only, but not the remaining two parties: Firm 2 and the consumers.

Conclusions

In this study, we use a nested logit framework to model the aggregate demand of insecticide firms in India by using monthly sales data. In doing so, we control for unobserved consumer heterogeneity by using the random coefficient specification, and also account for the potential endogeneity in price and distribution variables via the control function approach. Next, we estimate a supply-side price and distribution competition model taking the estimated demand model as an input. Methodologically, we illustrate how to estimate competitive interactions among firms in a multi-product form and multi-channel setting. Since most emerging economies' retail industries are composed of multiple firms, multiple product forms, and multiple channels, acknowledging this set-up is a factual requirement.

From our demand estimation, we find that the retail distribution affects the market demand in two ways: the direct effect – the higher the distribution level, the easier the transportation for consumers, and thus the higher the demand; and the indirect effect – the higher the brand's distribution level, the higher the market's price sensitivity for that brand. We further find that ignoring distribution in the demand estimation significantly biases the market's response to firms' pricing decisions. From our supply-side estimation, we find that there are significant differences in the estimated production (distribution) costs across firms, product forms, (retail channels), and years (to a lesser extent). We also find that ignoring distribution and solely focusing on price optimization, as previous studies have done, results in overestimating the equilibrium manufacturer profit margins by 7% to 55%. Last, we find that observed marketing mix variations can be better explained by firm competition rather than by tacit collusion.

Based on our estimation results, we conduct a series of simulation studies to examine the policy implications of our distribution and price competition model. Our simulation studies show that understanding the asymmetries in consumers' preferences for brands, product forms, and retail channels, as well as firms' distribution costs, are important in understanding firms' reactions to changes in the emerging marketplace. Our first study shows that as the transportation infrastructure improves (due to the construction of new roads and highways) in the emerging marketplace, firms should respond by reallocating their distribution resources asymmetrically – both firms (only the more preferred firm) should increase their (its) presence in the more (less) preferred paan-plus (general) stores channel, rather than changing their prices. Our second study shows that the more preferred firm's exclusive distribution 1) enables that firm to charge higher prices and 2) causes both firms to shift their distribution from the more preferred and costlier paan-plus stores channel to the general stores channel.

To reiterate, our study makes multiple contributions to marketing academia and practice as this is the first study to a) model competitive interactions in emerging market firms across two important marketing decisions, i.e. distribution and price (using a unique granular measure of distribution intensity), b) model marketing interactions at the multi-brand, multi- product form, and multi-channel levels, and c) guide emerging market managers to make profitable marketing-mix decisions by providing an approach to respond to changes in the emerging marketplace.

Although we provide significant contributions to the marketing literature and practice, our study faces some limitations. It is worth noting that our findings are specific to emerging markets and may not be generalizable to mature markets for the following reasons. First, the unavailability of products due to limited, expensive, and unorganized retail distribution is a major issue for emerging market manufacturers; whereas in mature markets, firms may not typically face these issues due to their organized and streamlined retail distribution strategies. Second, in most emerging markets, firms use multiple channels/store formats (e.g., mom-and-pop stores, general stores, etc.) to reach their end-consumers, unlike firms in mature markets which distribute their products through big-super-hyper-markets. Further, these retail channels are typically unique to the corresponding emerging market setting (for example, kirana and general stores in the Indian market; professional and specialty stores in the Chinese market[46]) and are significantly different from the retail channels in mature markets. Lastly, shelf spaces in a retail channel might be more limited in emerging markets (compared to mature markets) due to smaller store sizes and existing physical/locational constraints (such as lack of storage facilities, proper air conditioning etc.). Although manufacturers in mature markets also face limited shelf space issues, most major manufacturers manage to acquire required shelf spaces for their products. Given such differences, insights from our study may not be applicable to developed market managers.

One of the limitations of our study is that, since our focus is on distribution competition and price, we treat the effects of other unobserved marketing variables (such as advertising) as exogenous demand shocks. Although we estimate the demand and supply side with two marketing decisions, investigating how firms compete with other marketing mix instruments is an important issue for future research, permitting that the data is available. Moreover, if the data (of a firm's new versus already existing distributing stores) is available, modeling fixed costs of distribution can be another fruitful research direction. Second, our data is at the national level and as a result, we are not able to model the regional heterogeneity (some regions have only one kind of store and/or one form of product). Future research may capture such an important decision. Finally, there can be potential overlap among the channels of a firm and also some overlap with the competitors' channels. Although we do not observe such overlap, modeling the impact of such overlap upon the availability of data can bring new insights.

[46] http://www.funggroup.com/eng/knowledge/research/china_dis_issue114.pdf

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Figure 1: Nested Logit Tree

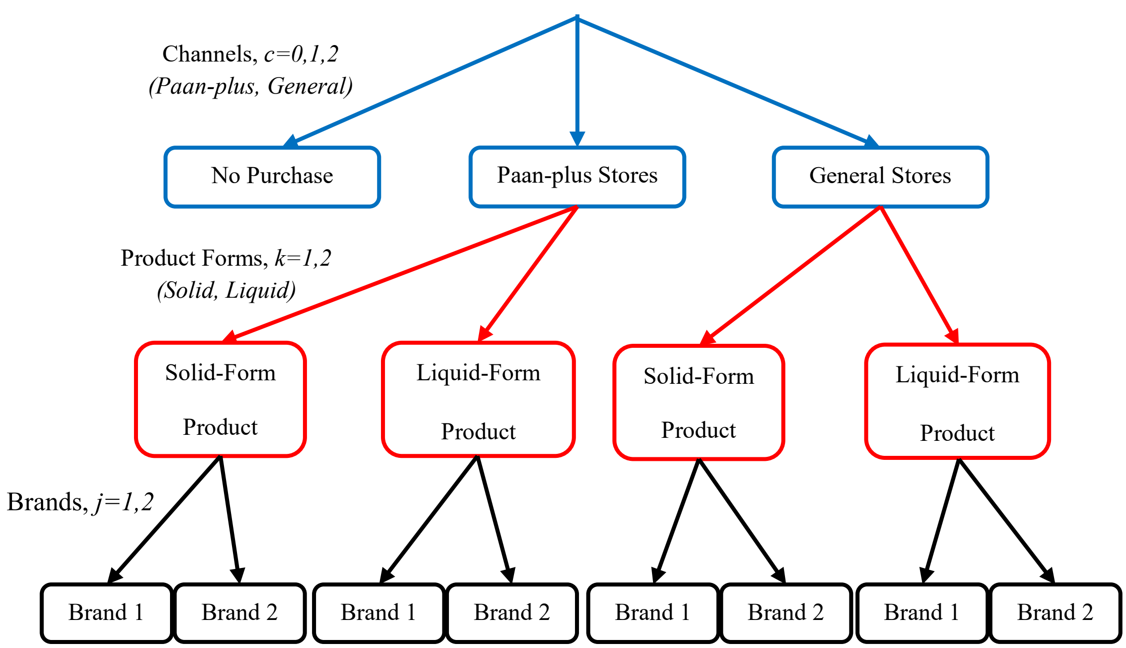

Figure 2: Equilibrium Prices, Distribution Levels, Profits, and Surplus versus Percentage Decrease in Distribution Costs

Figure 2A: Prices versus Percentage Decrease in Distribution Cost

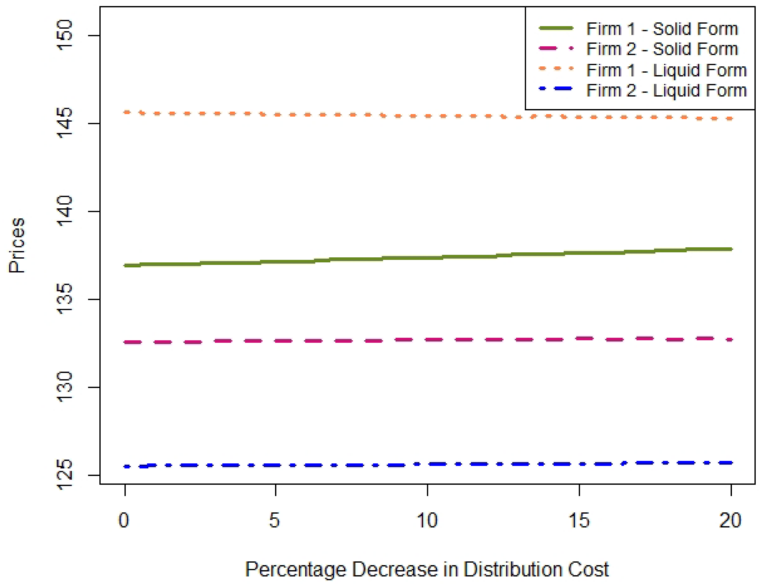

Figure 2B: Distribution Levels versus Percentage Decrease in Distribution Cost

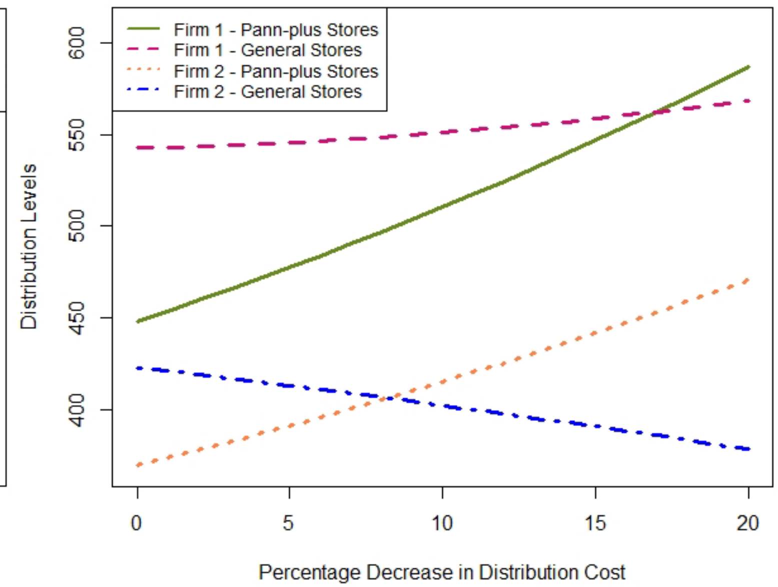

Figure 2C: Profits versus Percentage Decrease in Distribution Cost

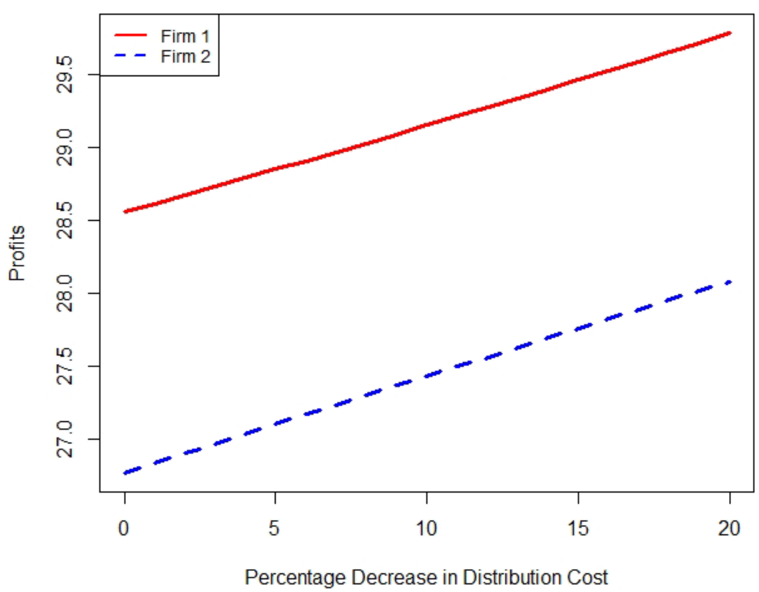

Figure 2D: Consumer Surplus versus Percentage Decrease in Distribution Cost

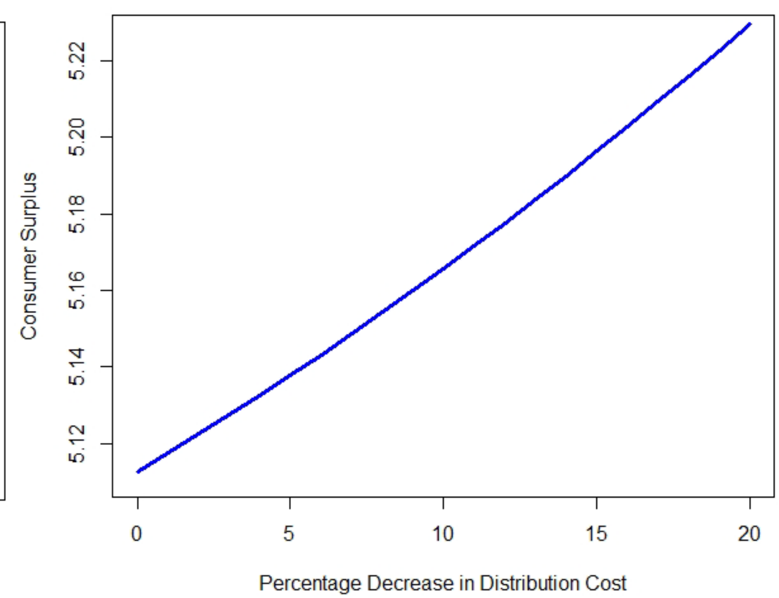

Figure 3: Equilibrium Prices, Distribution Levels, Profits, and Surplus versus Exclusivity Percentage of Firm 1

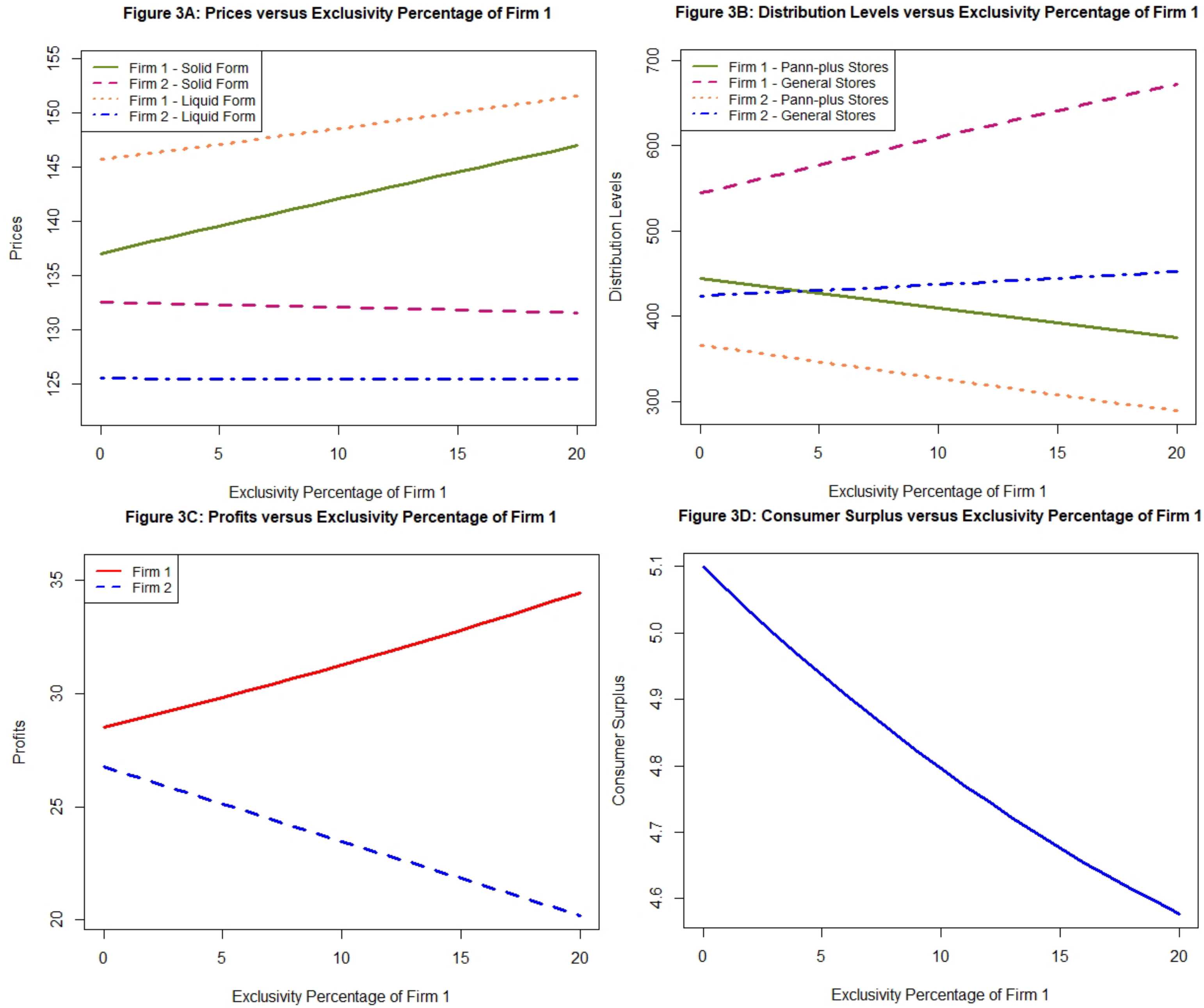

Table 1: Related Literature and Positioning of the Current Study

	Manufacturer Interactions	Pricing	Multi-channel Distribution	Market	Multi-channel-multi-product form setting
Rangan and Jaikumar 1991	No	Yes	Yes	Developed	No
Trivedi 1998	Yes	Yes	Yes	NA*	No
Balasubramanian 1998	No	Yes	Yes	NA	No
Sharma and Mehrotra 2007	No	No	Yes	NA	No
Brynjolfsson et al. 2009	No	No	Yes	Developed	No
Jill Avery et al. 2012	No	No	Yes	Developed	No
Homburg et al. 2014	No	No	Yes	Developed	No
Cao and Li 2015	No	No	Yes	Developed	No
Pauwels and Neslin 2015	No	No	Yes	Developed	No
This Study	**Yes**	**Yes**	**Yes**	**Emerging**	**Yes**

*NA represents Not-Applicable

Table 2: Descriptive Statistics

	Mean	Standard Dev.
Price of Firm 1 in Solid Product Form	136	14
Price of Firm 1 in Liquid Product Form	147	2
Price of Firm 2 in Solid Product Form	132	9
Price of Firm 2 in Liquid Product Form	126	13
Distribution of Firm 1 in Paan-plus Stores (in raw Numbers)	399	35
Distribution of Firm 1 in General Stores (in raw Numbers)	502	19
Distribution of Firm 2 in Paan-plus Stores (in raw Numbers)	356	50
Distribution of Firm 2 in General Stores (in raw Numbers)	482	60
Expected Population Each in Paan-plus Store Serves	2,353	523
Expected Population Each General Store Serves	11,986	849
Sales of Firm 1 in Paan-plus Stores	227	88
Sales of Firm 1 in General Stores	233	59
Sales of Firm 2 in Paan-plus Stores	258	78
Sales of Firm 2 in General Stores	328	49

Price in Paisa (Rupee/100), Distribution in thousands, Sales in hundred-thousands

Table 3: Demand-Side Parameter Estimates

Parameters	Model 1: Price and Distribution – MNL	Model 2: Price Only – NL	Model 3: Price and Distribution – NL
$\alpha_{\text{Firm=1 \| Solid Product Form, Paan-plus Stores}}$	1.361876***	1.798967***	1.807700***
$\alpha_{\text{Firm=1 \| Solid Product Form, General Stores}}$	-0.385638***	-2.378411***	0.802631***
$\alpha_{\text{Firm=1 \| Liquid Product Form, Paan-plus Stores}}$	-0.176789***	-17.070201***	0.566237***
$\alpha_{\text{Firm=1 \| Liquid Product Form, General Stores}}$	-0.188876***	-12.363232***	0.173262***
$\alpha_{\text{Firm=2 \| Solid Product Form, Paan-plus Stores}}$	0.989635***	1.459682***	1.089735***
$\alpha_{\text{Firm=2 \| Solid Product Form, General Stores}}$	0.890989***	-2.316726***	0.166078***
$\alpha_{\text{Firm=2 \| Liquid Product Form, Paan-plus Stores}}$	-2.924591***	-20.250492***	-0.992854***
$\alpha_{\text{Firm=2 \| Liquid Product Form, General Stores}}$	-1.912438***	-18.899213***	-2.221228***
β_{Price}	-0.044604***	-0.032315***	-0.040079***
$\beta_{\text{Distribution}}$	0.003613***		0.003902***
$\beta_{\text{Price x Distribution}}$	-0.000009***		-0.000011***
θ_{Summer}	-0.094805***	-0.083908***	-0.096157***
$\theta_{\text{Minimum Temperature}}$	0.015249***	0.014024***	0.013606***

$\theta_{\text{Maximum Temperature}}$	-0.040662***	-0.033255***	-0.038679***
θ_{Rainfall}	-0.000082***	-0.000055***	-0.000114***
$\varphi_{\text{Price Residual - Firm =1 \| Solid Product Form}}$	0.002997***	0.010144***	0.002590***
φ Price Residual - Firm =1 | Liquid Product Form	0.015546***	0.030652***	0.009889***
φ Price Residual - Firm =2 | Solid Product Form	0.009583***	0.007989***	0.007781***
φ Price Residual - Firm =2 | Liquid Product Form	0.023080***	0.030914***	0.018214***
φ Distribution Residual - Firm =1 | Solid Product Form, Paan-plus Stores	0.013900***		0.011222***
φ Distribution Residual - Firm =1 | Solid Product Form, General Stores	-0.000663***		0.000873***
φ Distribution Residual - Firm =1 | Liquid Product Form, Paan-plus Stores	0.018772***		0.021857***
φ Distribution Residual - Firm =1 | Liquid Product Form, General Stores	0.006897***		0.006223***
φ Distribution Residual - Firm =2 | Solid Product Form, Paan-plus Stores	0.000168***		-0.000959***
φ Distribution Residual - Firm =2 | Solid Product Form, General Stores	-0.002241***		-0.001109***
φ Distribution Residual - Firm =2 | Liquid Product Form, Paan-plus Stores	0.579555***		0.620739***
φ Distribution Residual - Firm =2 | Liquid Product Form, General Stores	0.058026***		0.059159***
$\sigma_{\text{Firm=1, Solid Product Form, Paan-plus Stores}}$	1.307344***	0.443141***	0.658274***
$\sigma_{\text{Firm=1, Solid Product Form, General Stores}}$	0.440758***	2.135500***	0.730913***
$\sigma_{\text{Firm=1, Liquid Product Form, Paan-plus Stores}}$	0.714133***	0.016816***	3.092251***
$\sigma_{\text{Firm=1, Liquid Product Form, General Stores}}$	0.080903***	2.658132***	0.246824***
$\sigma_{\text{Firm=2, Solid Product Form, Paan-plus Stores}}$	1.116059***	0.785814***	1.378974***
$\sigma_{\text{Firm=2, Solid Product Form, General Stores}}$	2.673028***	1.756758***	0.859093***
$\sigma_{\text{Firm=2, Liquid Product Form, Paan-plus Stores}}$	0.053846***	3.241646***	0.630437***
$\sigma_{\text{Firm=2, Liquid Product Form, General Stores}}$	1.624261***	14.287679***	0.028899***
σ_{Price}	0.013012***	0.008780***	0.010421***
$\sigma_{\text{Distribution}}$	0.002008***		0.000334***
$\lambda_{\text{Solid Product Form\|Paan-plus Stores}}$		1.345457***	1.057025***
$\lambda_{\text{Solid Product Form\|General Stores}}$		1.009945***	3.074303***
$\lambda_{\text{Liquid Product Form\|Paan-plus Stores}}$		0.215446***	2.612888***
$\lambda_{\text{Liquid Product Form\|General Stores}}$		0.495955***	0.986200***
$\lambda_{\text{Paan-plus Stores}}$		0.981549***	1.531355***
$\lambda_{\text{General Stores}}$		0.379499***	1.146316***
Log-Likelihood	20,046.4M	20,056.5M	20,045.1M
BIC	40,092.7M	40,113.0M	40,090.2M

***significant at 1%|**significant at 5%|*significant at 10%

Table 4: Supply-Side Parameter Estimates - Marginal Cost Estimation

Variables	Model 1: Bertrand Price and Distribution Competition	Model 2: Price Only Bertrand Competition	Model 3: Collusion in Pricing
Intercept	92.7451***	67.72348***	58.9702***
Dum11 (Firm 1, Solid Form)	-32.64***	-14.7355***	-30.9848***
Dum21 (Firm 2, Solid Form)	-18.9754***	-7.32721***	17.4403***
Dum12 (Firm 1, Liquid Form)	3.0917***	-0.0075	10.528***
Year 1	-14.7265***	-12.6933***	0.3382
Year 2	-18.977***	-14.2222***	-17.114***
Year 3	-17.5479***	-12.4049***	-13.6406***
Dum11*Year1	34.8939***	34.5135***	35.7045***
Dum11*Year2	44.0107***	41.59668***	50.4865***
Dum11*Year3	19.0766***	14.09259***	20.4317***
Dum12*Year1	26.0565***	21.76817***	-3.0698
Dum12*Year2	33.2727***	26.54989***	9.3306**
Dum12*Year3	17.5908***	12.18747***	17.5013***
Dum21*Year1	12.5506***	21.40931***	14.4409***
Dum21*Year2	18.0411***	22.76995***	26.6711***
Dum21*Year3	17.5239***	21.59883***	18.1615***

R^2	95.97%	92.09%	87.78%
Adj. R^2	95.61%	91.39%	86.68%
SSR	1,002.24	1,421.90	7,026.77

***significant at 1%|**significant at 5%|*significant at 10%

Table 5: Supply Side Parameter Estimates - Distribution Cost Estimation

Variables	Model 1: Bertrand Price and Distribution Competition	Model 2: Collusion in Distribution
Intercept	4022.8***	4960.6***
D111 (Firm 1, Solid Form, Paan-plus stores)	85893.6***	64263.4***
D112 (Firm 1, Solid Form, General stores)	2852.4**	15951.4***
D121 (Firm 1, Liquid Form, Paan-plus stores)	8482.4***	33218***
D122 (Firm 1, Liquid Form, General stores)	139	-728
D211 (Firm 2, Solid Form, Paan-plus stores)	59683***	1350.7
D212 (Firm 2, Solid Form, General stores)	485.7	428.7
D221 (Firm 2, Liquid Form, Paan-plus stores)	1782.2	11068.2***
Year 1	-412	270.9
Year 2	-1056.1	-210.8
Year 3	-123.3	-380.6
D111 * Year 1	-42836.9***	-20806***
D111 * Year 2	-46039.6***	-23985.5***
D111 * Year 3	-4449.3***	-3231
D112 * Year 1	-5109.3***	-14598.4***
D112 * Year 2	-4187.3**	-13072***
D112 * Year 3	-429.6	-333.8
D121 * Year 1	7457.6***	2571.9
D121 * Year 2	6252.7***	4130.7*
D121 * Year 3	1319.8	469.6
D122 * Year 1	732.2	408.1
D122 * Year 2	1273.9	753.8
D122 * Year 3	302.1	551.4
D211 * Year 1	-11659.1***	1754.9
D211 * Year 2	-19784.5***	1334.6
D211 * Year 3	-728.6	144
D212 * Year 1	-2120.7	-587.5
D212 * Year 2	-1240.1	913.3
D212 * Year 3	121.4	309.2
D221 * Year 1	20734.2***	10650.8***
D221 * Year 2	9355.2***	5382**
D221 * Year 3	5206.1***	5932.2***
R^2	98.86%	96.46%
Adj. R^2	98.75%	96.13%
SSR	2,752.9M	4,822.5M

***significant at 1%|**significant at 5%|*significant at 10%

Table 6: Calculated Equilibrium Profit Margins

	Price Only Model				Proposed Model			
Profit Margins in Paisa	Year 1	Year 2	Year 3	Year 4	Year 1	Year 2	Year 3	Year 4
Firm 1 - Solid Form	71.3	71.2	68.9	69.4	65.8	66.4	61.9	62.2
Firm 1 - Liquid Form	70.2	71.3	71.1	78.9	53.0	52.7	52.2	50.8
Firm 2- Solid Form	68.7	69.6	63.4	63.7	53.1	54.2	49.8	50.4
Firm 2 - Liquid Form	62.6	62.5	65.2	79.8	39.7	42.2	45.4	54.8